STAT 24410/30030, Homework 1

Due Thursday, October 10, at 2:00pm

0. (0 pts) Read Chapter 1 of the Lecture Notes!

1. (6 pts) Lecture Notes, Chapter 1, Exercise 1.

2. (6 pts) Lecture Notes, Chapter 1, Exercise 2.

3. (6 pts) Lecture Notes, Chapter 1, Exercise 3.

4. (8 pts) Lecture Notes, Chapter 1, Exercise 4.

5. (10 pts) Lecture Notes, Chapter 1, Exercise 5.

6. (10 pts) Lecture Notes, Chapter 1, Exercise 6.

7. (8 pts) Lecture Notes, Chapter 1, Exercise 7.

8. (7 pts) Lecture Notes, Chapter 1, Exercise 8.

9. (9 pts) Lecture Notes, Chapter 1, Exercise 9.

10. (10 pts) Let X be a random variable with cumulative distribution function F. Find the cumulative distribution functions of $|X|$ and X^+, where X^+ is equal to X if $X > 0$ and is equal to 0 otherwise.

11. (10 pts) Suppose that $X_1, ..., X_n$ are independent and that each has cumulative distribution function F and let

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} I_{\{X_i \leq x\}}, \quad x \in R.$$

(a) For a given realization of $X_1, ..., X_n$, find a random variable for which F_n is its cumulative distribution function.

(b) What happens to $F_n(x)$ as n goes to infinity?

12. (10 pts) In this exercise, we will use simulations to empirically understand some of our findings above. **Provide all code you use.**

(a) The goal here is construct an approximation for the cumulative distribution function for X^+ (defined in Exercise 10) where X is distributed $N(-1, 1)$. Simulate a sample $x_1, ..., x_n$ of size $n = 20$ from the $N(-1, 1)$ distribution. On the same plot show the cumulative distribution of X^+ (using the formula you derived in Exercise 10) and an approximation based on the data (see Exercise 11) - the approximation should be calculated on a grid of points. Repeat this for $n = 50$ and $n = 100$ and comment on the three plots.

(b) Let X be distributed according to a standard normal and $Y = \sin(X)$. Use simulations to draw an approximation of the cumulative distribution function of Y. **Optional:** think about the mathematical derivation of this cumulative distribution function.